DRDC September 2024 Update

Initial Simulation Results and Next Steps

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System Model

- We consider an MFR that can:
  - **search:** detect previously undetected targets, or
  - **track:** update known targets
- Surveillance region is quantized into a set of N fixed non-overlapping cells
- The radar uses a number of beams $\boldsymbol{A} = \{1, \ldots, A\}$, between which it must divide its time.
  - time allocated to a beam to perform a task denoted as τ_a
- Initially, we assume a scan duration ΔT of 1 second.
  - Surveillance tasks have a norminal rate of once per ΔT

JAX

- JAX is a Python library for numerical computing

Demo